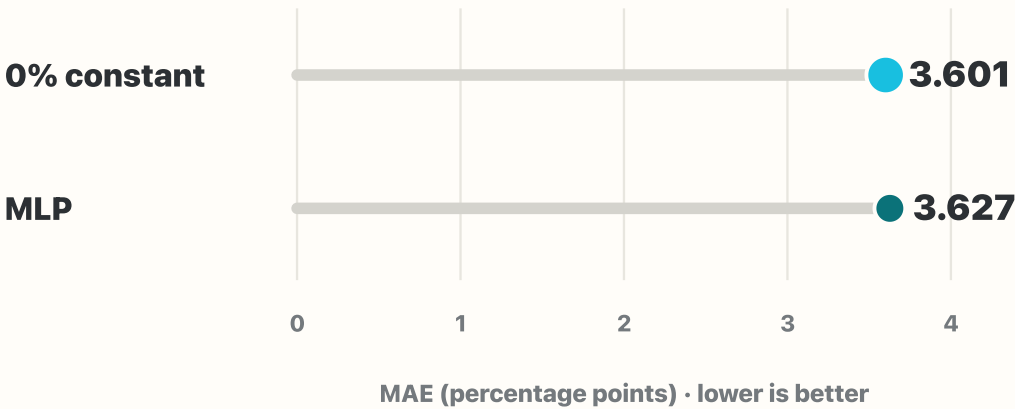
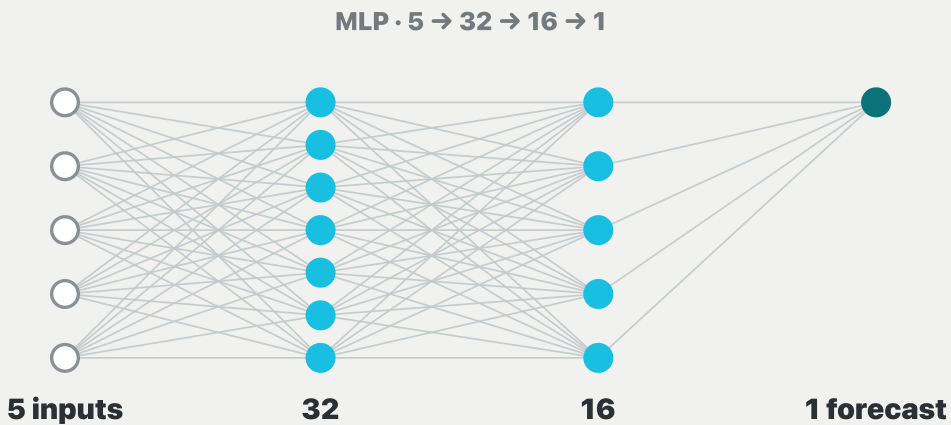


# I SPENT 24 HOURS ON A NEURAL NETWORK. ALL I GOT WAS THIS LOUSY T-SHIRT.

My fancy model can learn very complex functions.  
"Predict 0% growth" still beat it on mean absolute error (MAE).

EXPLANATION OF THE TARGET VARIABLE  
NEXT-QUARTER DOMESTIC DEPOSIT GROWTH

|                |                     |
|----------------|---------------------|
| Deposits       | \$100m → \$96m      |
| Actual growth  | -4%                 |
| Zero forecast  | 0%                  |
| Absolute error | 4 percentage points |



THE CONSTANT PREDICTED 0% GROWTH FOR EVERY BANK.

# THE BANK CAN LEND OUT MONEY YOU DEPOSIT. IT DOESN'T KEEP IT ALL IN CASH.

WHEN YOU WITHDRAW, SOME FUNDS MAY STILL BE TIED UP IN LOANS.



CASH IN THE VAULT

**\$10**



MONEY ALREADY OUT AS LOANS

**\$90**

A customer withdraws \$15. Cash is \$5 short.

**THE BANK'S TREASURY TEAM MUST FIND THE CASH.**

## HOW MUCH CASH IS ENOUGH?



Required reserve ratio: 0%. That does not mean unlimited lending.

There is no universal “keep 10%, lend 90%” rule. Capital, liquidity and other limits constrain lending.

The full liquidity coverage rule requires eligible liquid assets covering **100% of 30-day stressed net cash outflows.**

CASH

DEPOSITS

LOANS

CAPITAL

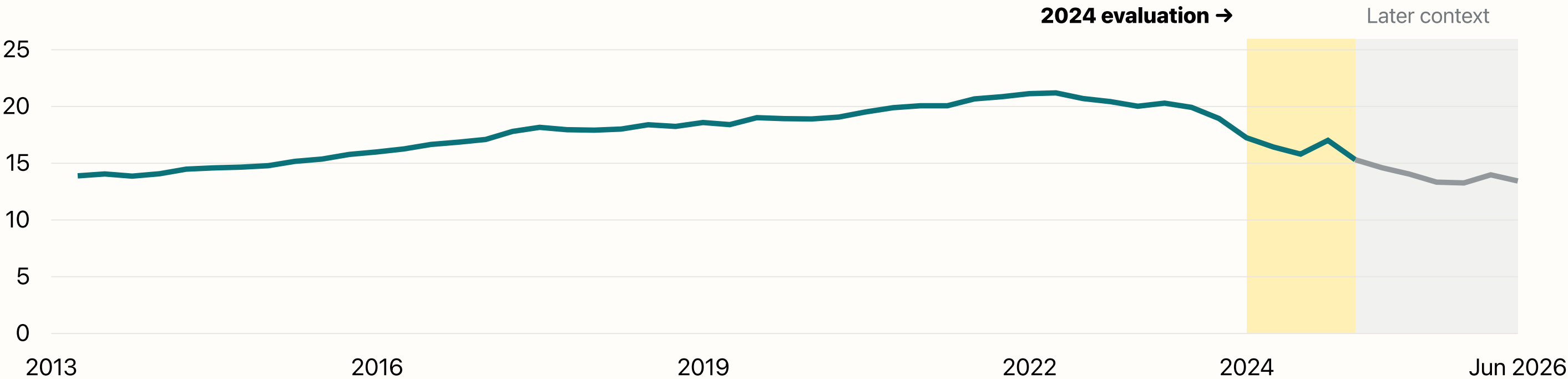
**CUSTOMERS DEPOSIT AND WITHDRAW MONEY.  
I'M FORECASTING THE NET CHANGE NEXT QUARTER.**

Example only. Reserve requirement ratios are 0%; separate liquidity rules vary by bank. Liquid assets include eligible securities, not just cash. Source: Federal Reserve, Regulation WW. This project forecasts deposit growth, not bank failure.

# "PREDICT 0%" WON. IS THIS THE GRAPH YOU WERE EXPECTING?

2013–2026 · ONE-BANK ILLUSTRATION. Zero won MAE across all test banks, not necessarily this bank.

YOUNG AMERICANS BANK · DOMESTIC DEPOSITS (\$ MILLIONS)



**Think of a piggy bank.**

It has \$100 today. "0% growth" guesses \$100 next quarter.  
If it ends up with \$105, the next guess starts at \$105.

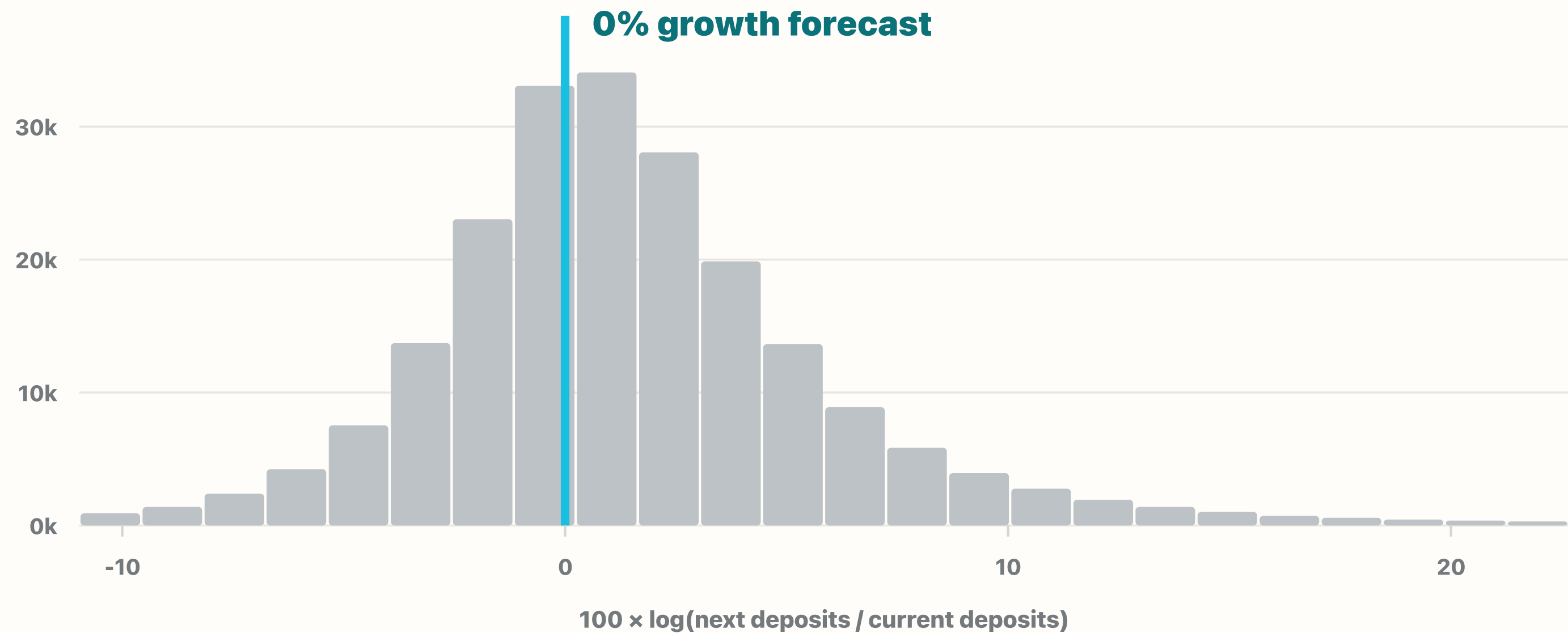
**The guess starts fresh every quarter.**

The balance can climb and fall over the years.  
We're guessing just **one quarter ahead**.

**So how big are those one-quarter changes across all the banks? →**

FDIC BankFind · CERT 27010 · 54 quarterly reports shown, Mar 2013–Jun 2026 · retrieved Sep 22, 2026. Piggy-bank amounts are illustrative.  
2013 is the project's data start. Gold marks the reused 2024 evaluation year; gray reports came later. Zero won MAE among the original four models.

# SO WHY WAS 0% GROWTH SO HARD TO BEAT?



**A LOT OF THE TRAINING TARGETS LIVE NEAR ZERO.**

Central 98% shown · full tails stayed in the experiment · display scale is 100 × log growth

# BIG ERRORS GET **SQUARED**. SO THEY CAN PULL THE MODELS AWAY FROM 0%.

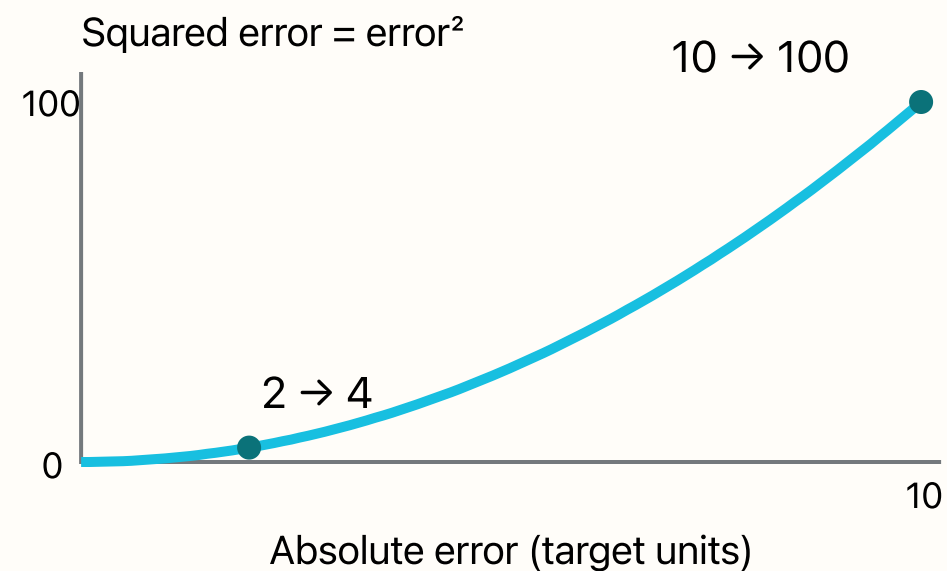
An error 5× as large gets 25× the squared-error penalty.

## FIXED BENCHMARK

### 0% BASELINE

$$\hat{y} = 0$$

Always predicts 0%.  
It never adjusts to reduce an error.



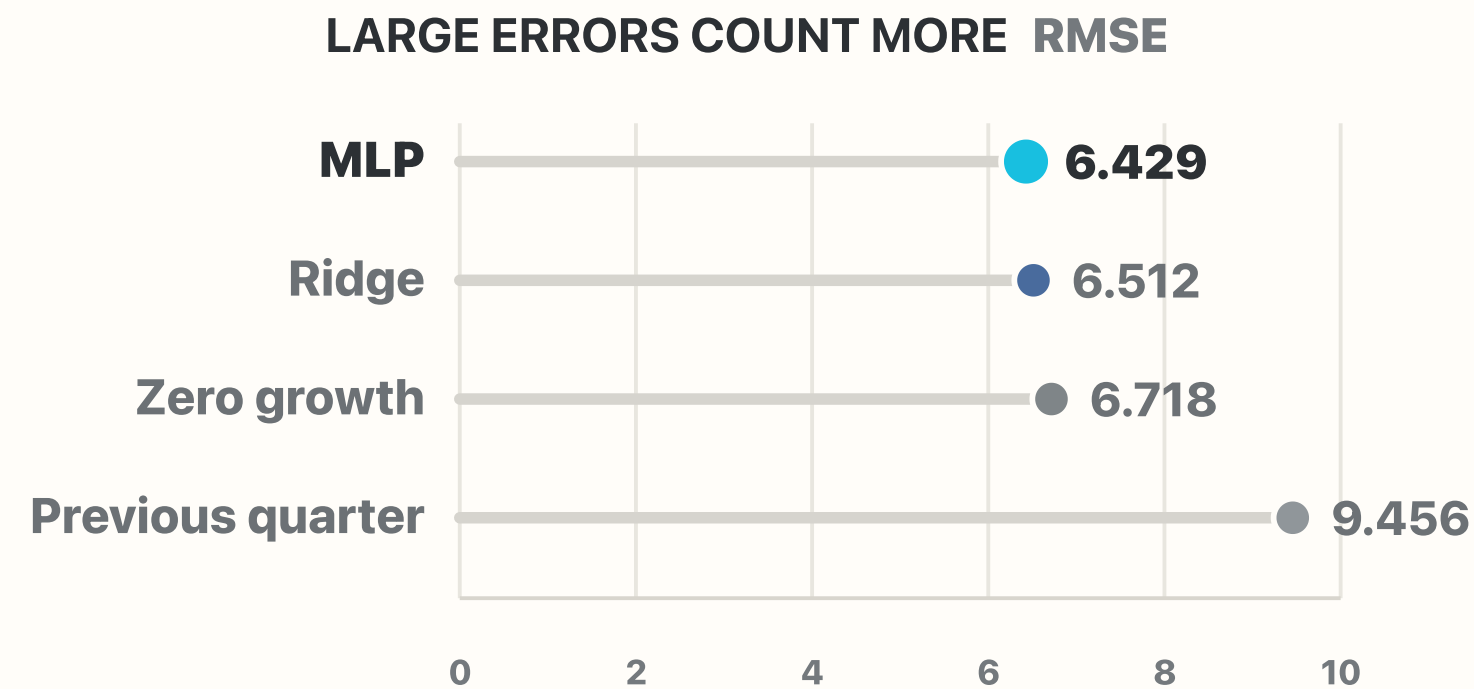
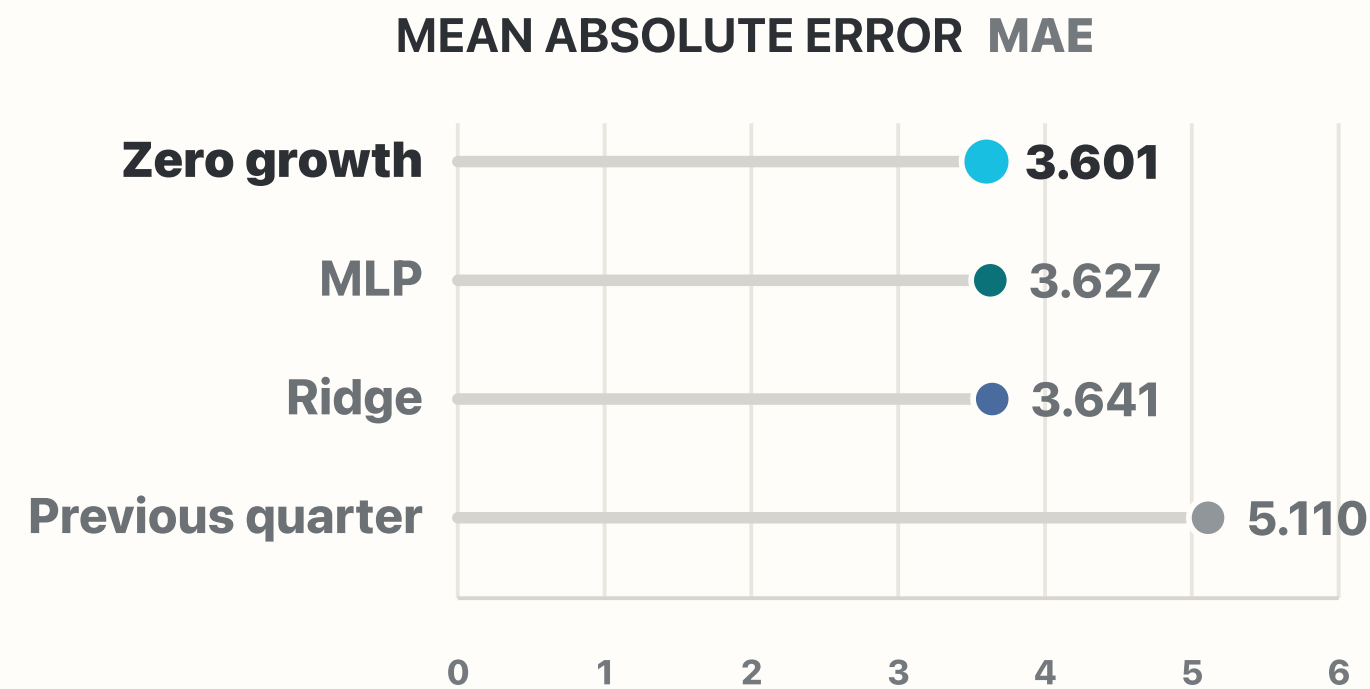
## LEARNED MODELS

### RIDGE + NEURAL NET

Can move away from 0% to reduce those expensive large training errors.

**A LOWER SQUARED ERROR CAN COME WITH  
A SLIGHTLY HIGHER MEAN ABSOLUTE ERROR.**

# ZERO WON MAE. THE NEURAL NETWORK WON RMSE.



**“WINNER” DEPENDS ON THE ERROR METRIC WE CHOOSE.**  
MAE averages absolute errors. RMSE squares errors first, so large errors count more.

# 0% TIES EVERY BANK.

## THE LEARNED MODELS TELL ME WHO TO REVIEW FIRST.

★ I am not a financial advisor. I do have a sorted table.

In this March 2024 queue, review Young Americans Bank first: Ridge predicts the largest deposit decline, −4.02%. Check its reports to understand the forecast; this is not a current distress warning.

ORDERED REVIEW QUEUE · SIMPLE LINEAR MODEL (RIDGE) · FIRST 10 OF 454

| RANK | BANK                       | PREDICTED GROWTH | 0% BASELINE |
|------|----------------------------|------------------|-------------|
| 01   | YOUNG AMERICANS BANK       | -4.02%           | 0% · tied   |
| 02   | PIONEER COMMERCIAL BANK    | -3.32%           | 0% · tied   |
| 03   | SUMITOMO MITSUI TR BK U SA | -2.83%           | 0% · tied   |
| 04   | OLD GLORY BANK             | -2.75%           | 0% · tied   |
| 05   | TURTLE MOUNTAIN STATE BANK | -2.11%           | 0% · tied   |
| 06   | COLUMBUS STATE BANK        | -2.05%           | 0% · tied   |
| 07   | FDS BANK                   | -2.03%           | 0% · tied   |
| 08   | BOELUS STATE BANK          | -1.96%           | 0% · tied   |
| 09   | LEMONT NATIONAL BANK       | -1.80%           | 0% · tied   |
| 10   | CITIZENS NB OF CROSBYTON   | -1.75%           | 0% · tied   |

GOAL: FIND MORE LOWEST-GROWTH BANKS WITH THE SAME 454 REVIEWS.

HIGHER IS BETTER.

Random selection

45.4 expected

Linear model (Ridge)

112 found

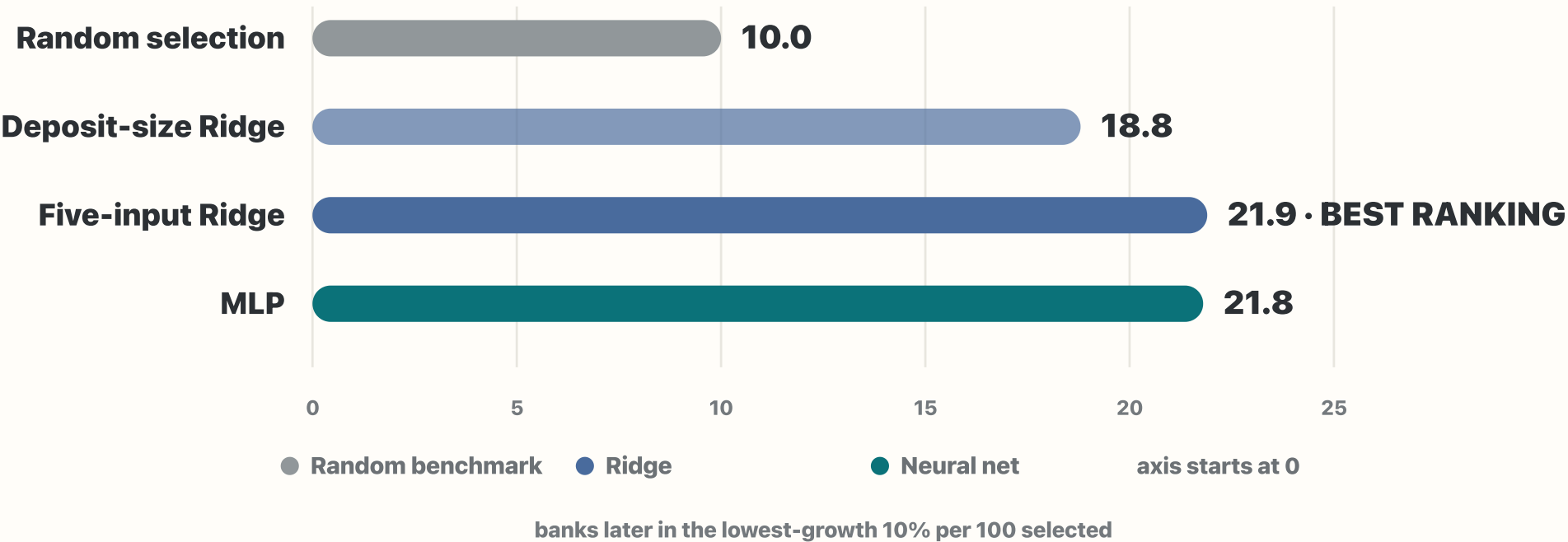
Neural network

109 found

4,536 eligible banks · March → June 2024. “Lowest-growth” = later in that quarter’s bottom 10% of deposit growth. 112 matches across the full linear-model queue; 109 for the neural net. Review priority is not a distress diagnosis.

# I THREW OUT 4 INPUTS. KEPT ONLY DEPOSIT SIZE.

I tested deposit size alone to see whether four extra inputs actually improved the review list.



**MAE WINNER: PREDICT 0% → 3.601.** Deposit-size Ridge: 3.654 · Five-input Ridge: 3.641 · Neural net: 3.627.

All three lost to zero on MAE. Zero ties every bank, so it cannot order a review list.

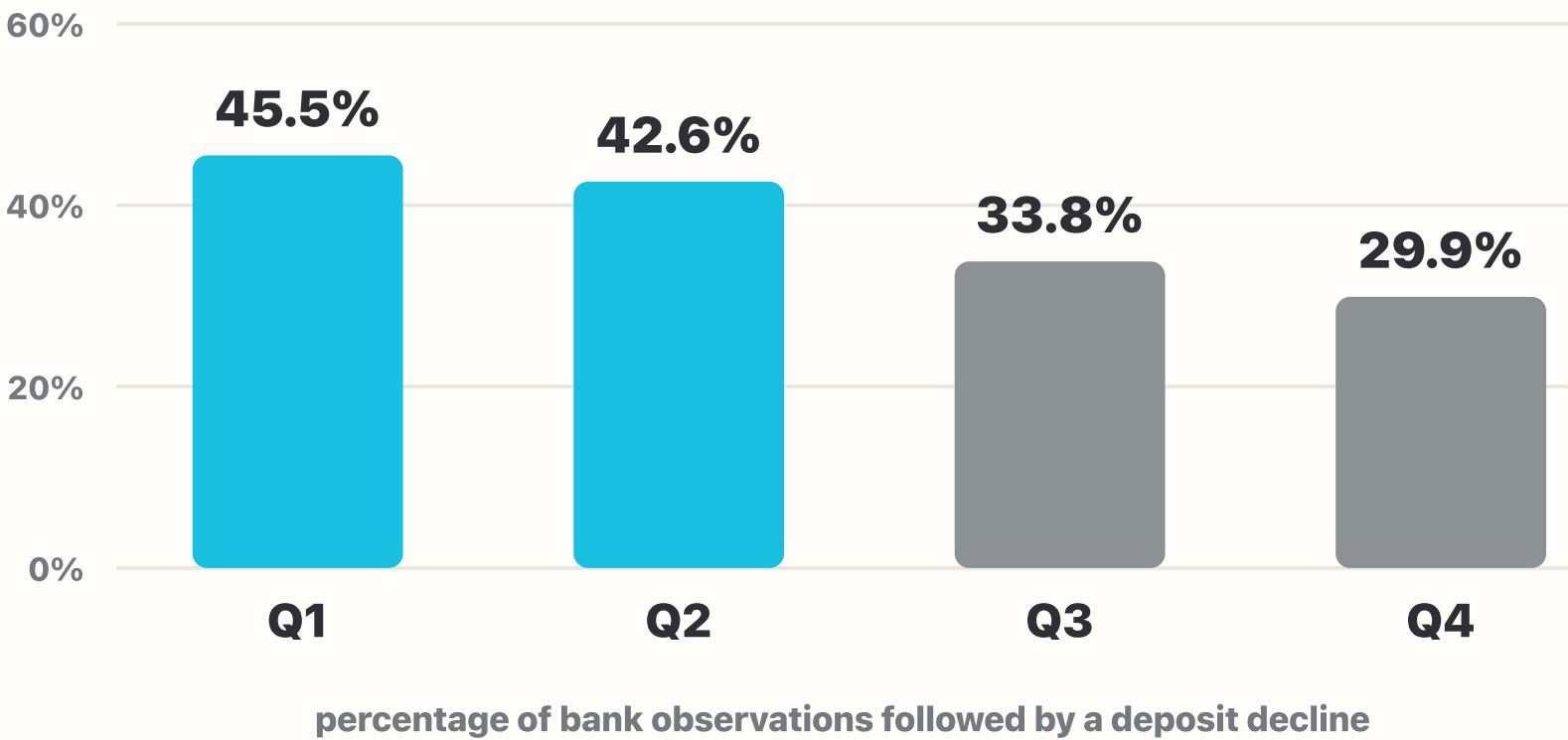
**DEPOSIT SIZE ALONE: 18.8 FOUND PER 100 REVIEWED.**  
**ALL FIVE INPUTS: 21.9. A GAP OF 3.1.**

# DEPOSIT DECLINES WERE MORE COMMON AFTER Q1 THAN Q4.

**Q1: 45.5%** of training bank-quarters were followed by a deposit decline. **Q4: 29.9%.**

That is how often declines happened, not “banks lost 45.5% of their deposits.”

HOW OFTEN DID A BANK’S DEPOSITS FALL NEXT QUARTER? · TRAINING YEARS 2013–2022

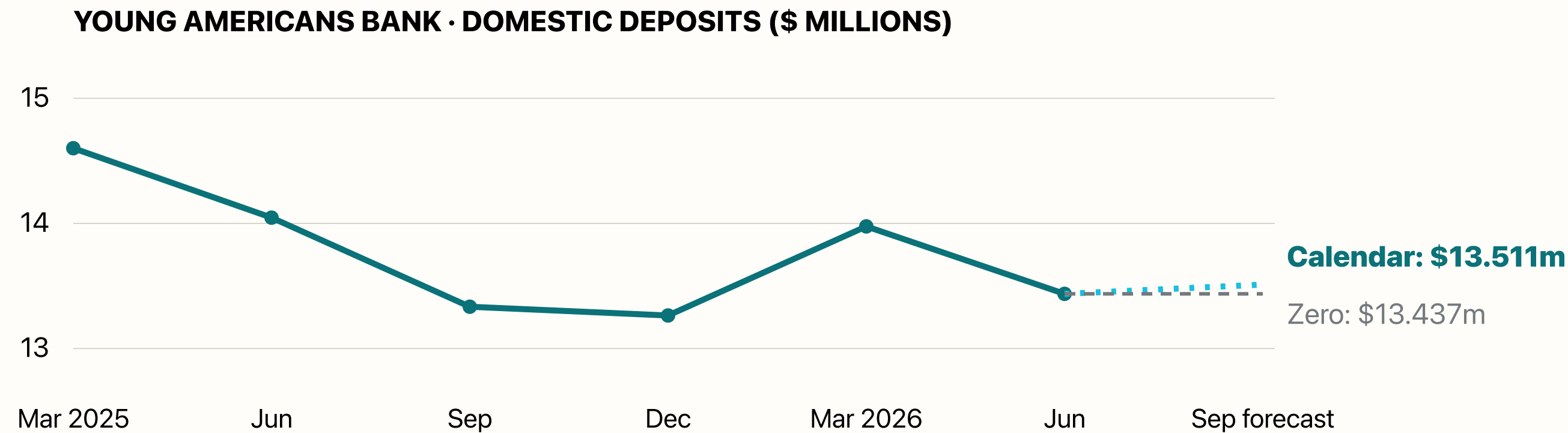


THE STARTING QUARTER MAY HELP SET A GROWTH FORECAST.

For each bar, count bank observations with a next-quarter decline, then divide by all observations starting in that quarter (training data through Sep 2022). Deposit dollars are not added together.

# MODEL CHOSEN FOR THE NEXT TEST: THE CALENDAR RULE. NOW WE WAIT FOR Q3 REPORTS.

Q3 ends September 30, 2026. Reports arrive afterward. The saved forecast is +0.550% per bank; zero is the benchmark.



Solid = reported balances. Dotted = calendar forecast (+0.550%). Dashed = no-change forecast.

**WHEN SEPTEMBER REPORTS ARRIVE, COMPARE MEAN ABSOLUTE ERROR.  
SAME BANKS. SAME QUARTER. LOWER MAE WINS.**

FDIC BankFind, CERT 27010, retrieved Sep 22, 2026. One-bank illustration of the saved Q2 median rule (+0.550399842%); test MAE across matched banks. Rule developed after seeing 2024. Score matched banks with valid positive June and September domestic deposits; disclose missing reports and merger cases. Example is illustrative. One quarter is not lasting proof.